

Minor in Robotics

Minor Details

Minor name	Minor in Robotics
Eligibility batch(es)	Batch 2025 onwards

Table 1: Minor in Robotics — Requirements

Sl.	Course code	Course name	L-T-P-C	Credits
Compulsory Core Courses (both required)				
All 2 listed courses required; Counts toward minor credits				
1	AR-523	Robot Manipulators	—	3
2	AR-520	Design Practicum of Mechatronic Systems	—	3
Electives (pick 2)				
Choose 2 of 20; Counts toward minor credits				
3	AR-502	Advanced Design Practicum	3-0-2-4	4
4	AR-504	Robot Programming	2-0-2-3	3
5	AR-505	Principles of Robot Autonomy	3-0-0-3	3
6	AR-506	Cognitive Robotics	3-0-0-3	3
7	AR-507	Probabilistic Robotics	3-0-0-3	3
8	AR-508	Marine Robotics	3-0-0-3	3
9	AR-509	Deep Learning for Robotics	3-0-2-4	4
10	AR-510	Underactuated Robotics	3-0-0-3	3
11	AR-511	Autonomous Mobile Robots	3-0-0-3	3
12	AR-512	Rapid Prototyping and Tooling	3-0-2-4	4
13	AR-513	Unmanned Aerial Systems (UAS)	3-0-0-3	3
14	AR-514	Vision and Learning Based Control	3-0-0-3	3
15	AR-515	Sensors and State Estimation	3-0-0-3	3
16	AR-521	Control of Robotic Systems	—	3
17	AR-522	Robot Vision	—	3
18	AR-524	Robot Simulators	—	3
19	AR-525	Reinforcement Learning for Robotics	—	3
20	AR-526	Multimodal Learning	—	3
21	AR-527	Patenting Through Design Thinking	—	3
22	EE-555	Intelligent Control System	3-0-0-3	3

Requirement notes

- **Compulsory Core Courses (both required):** AR-523 Robot Manipulators + AR-520 Design of Mechatronic Systems (6 credits total).
- **Electives (pick 2):** Any 2 electives from the list (6 credits). New electives may be added by the Centre from time to time.